

Pruning Graph Convolutional Networks to select meaningful graph frequencies for fMRI decoding

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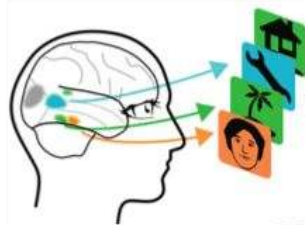
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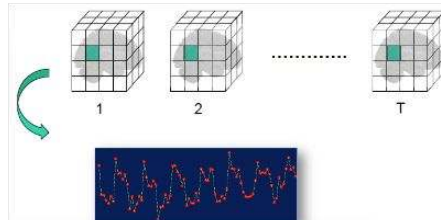


Context



Decoding
Extracting brain activity patterns that discriminate conditions and predict them from unseen neural activity

fMRI
Spatio temporal imagery technic that shows brain activity



Brain & Graphs
The brain can be parcelled in multiple regions that can be connected using graphs

Graph Signal Processing

- Let consider a graph $\mathcal{G} = \langle \mathcal{V}, \mathcal{E}, W \rangle$
 $W \in \mathbb{R}^{N \times N}$ its weight matrix
and $D \in \mathbb{R}^{N \times N}$ its degree matrix
- We can then compute its Laplacian : $L = D - W$
That can be diagonalized : $L = U \Lambda U^T$
- Let $x \in \mathbb{R}^N$ be a signal on $\mathcal{G}$
Its Graph Fourier Transform is given by :
$$\hat{x} = \text{GFT}(x) = U^T x$$

and its inverse by :
$$x = \text{GFT}^{-1}(\hat{x}) = U \hat{x}$$

Challenges

- Which graph frequencies are the most relevant to predict the experimental task from corresponding fMRI activation maps?
- Are these frequencies consistent for different underlying graphs (functional vs. structural) and deep learning architectures?
- Does selecting specific graph frequencies increase brain decoding performance?



Our contribution TL;DR

We propose a new methodology to perform graph frequency selection that we compare to standards methods :

- The core idea is to select by a learning process the appropriate graph frequencies within the signals.
- We consistently show its efficiency, robustness and replicability

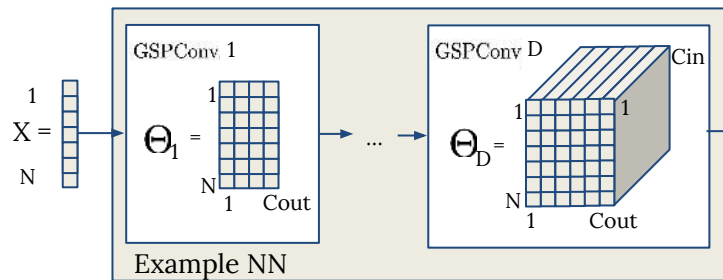


Focus on our pruning / frequency selection methodology

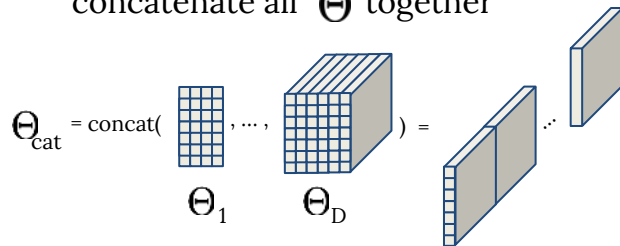
- We introduce a Deep Learning trainable layer that allows to convolute our signal over the graph :

$$\text{GSPConv}_{c_{in}, c_{out}}(\mathbf{X}) = \sigma \left(\text{GFT}^{-1} \left(\text{GFT}(\mathbf{X}) \otimes \Theta \right) \right) \quad \text{with } \Theta \in \mathbb{R}^{N \times c_{in} \times c_{out}} \quad (N : \text{number of frequencies})$$

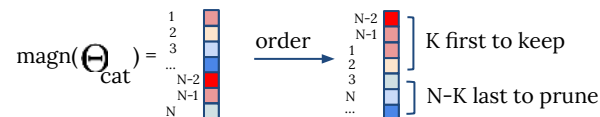
- We use this layer to build Neural Networks



- At each step of the training we concatenate all Θ together

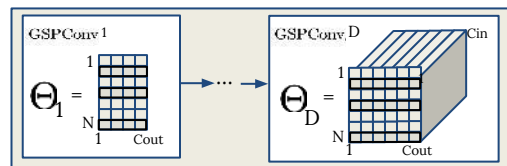


- To identify the frequency to prune

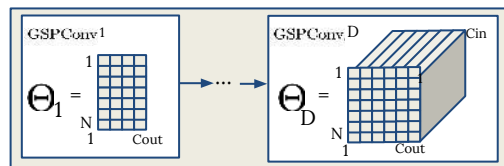


- Pruning phase

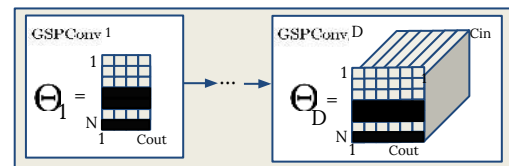
$$\alpha(s) = \alpha_{\min} \left(\frac{\alpha_{\max}}{\alpha_{\min}} \right)^{\frac{s}{s_{\text{final}}}}$$



step 0 : $\alpha = \alpha_{\min}$

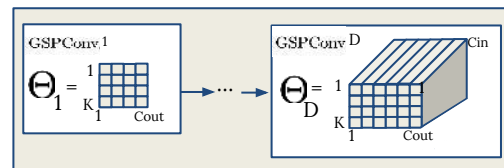


step s : $\alpha(s)$

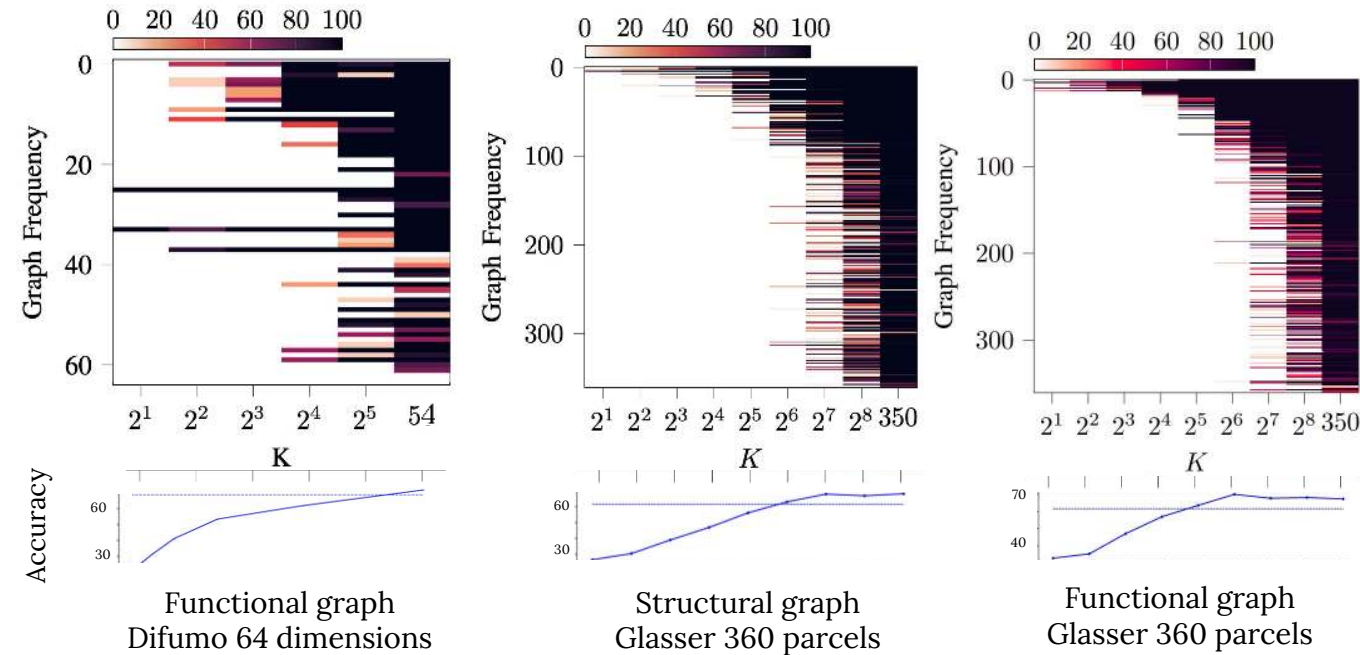


last step : $\alpha = \alpha_{\max}$

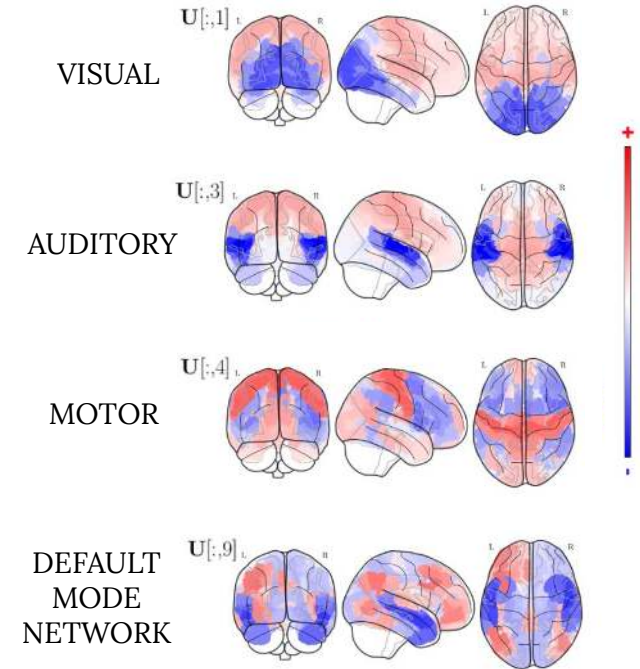
- LR-Rewinding phase



Results



Occurrences of each graph frequency for various fixed K
Corresponding accuracy (23 classes) below
(dotted line correspond to baseline without pruning)



Visualization of most frequently selected graph eigenvectors of the functional graph (Glasser 360) on the glass brain.

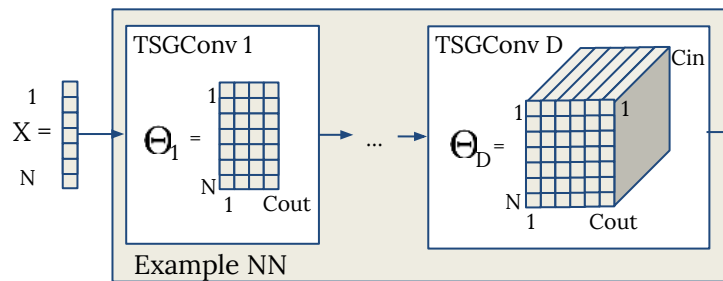
Conclusions & Perspectives

- ❖ Original use of a pruning methodology.
 - ❖ A posteriori validation of the use of GNN.
 - ❖ Only few frequencies are required to achieve good performances even among very divers tasks.
 - ❖ Our code is reusable and easy to adapt.
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- ❖ Use our methodology to identify spatial patterns characteristics of specific decoded task or of a patient profile.
 - ❖ Assess our methodology on other tasks/domains or for other purposes.

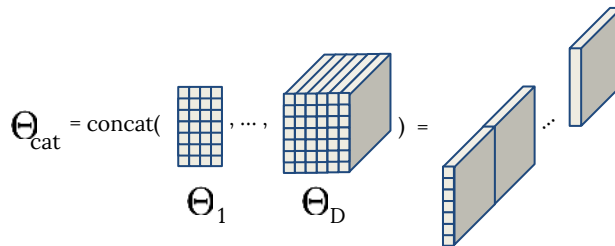


Focus on our pruning / frequency selection methodology

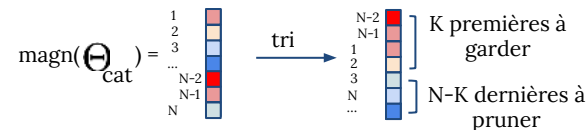
- On utilise la couche TSGConv pour construire des réseaux de neurones



- A chaque étape de l'entraînement, on concatène tous les Θ ensemble

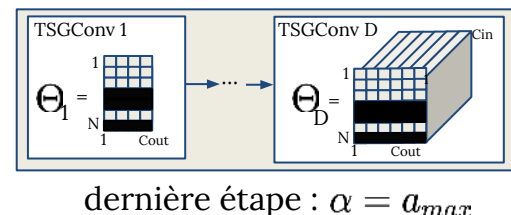
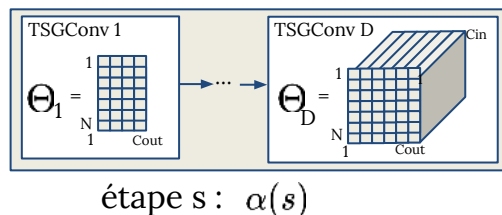
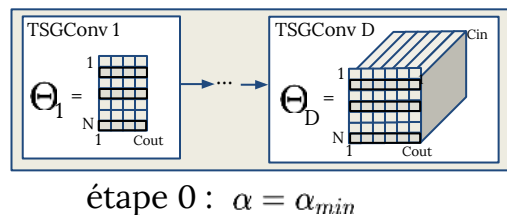


- Pour identifier les fréquences à élaguer



1) Étapes de pruning

$$\alpha(s) = \alpha_{\min} \left(\frac{\alpha_{\max}}{\alpha_{\min}} \right)^{\frac{s}{s_{\text{final}}}}$$



2) Étapes de LR-Rewinding

